

MedForge Study Guide — Acromegaly

Educational draft - source review required

Big picture

Acromegaly is a chronic endocrine disorder caused by hypersecretion of growth hormone (GH), most commonly due to a pituitary adenoma, leading to elevated insulin-like growth factor 1 (IGF-1) levels [S1]. It presents insidiously with progressive physical changes and systemic complications, including metabolic disturbances, cardiovascular disease, and increased cancer risk [S4]. Early diagnosis and treatment are critical due to high morbidity and mortality, especially in older patients [S1][S5]. The disease is often missed for years, contributing to poor outcomes [S2].

Core mechanism

GH hypersecretion drives increased IGF-1 production, which mediates many of the disease's effects. GH directly impairs insulin signaling in peripheral tissues, promotes lipolysis, and reduces glucose uptake, leading to insulin resistance independent of obesity [S3]. This results in a lean phenotype with reduced adipose tissue mass despite insulin resistance [S3]. The pathophysiology creates a unique metabolic state where diabetes mellitus is common, even in the absence of obesity [S3].

Anatomy/physiology/pathology/pharmacology connections when supported

The pituitary gland is the primary source of GH overproduction, typically via a pituitary adenoma [S1][S4]. Elevated GH and IGF-1 levels are central to diagnosis and monitoring [S4]. Biochemical confirmation requires elevated serum IGF-1 and lack of GH suppression after oral glucose administration [S4]. Imaging, particularly pituitary MRI, is advised to identify the underlying adenoma [S4].

Treatment options include transsphenoidal surgery (first-line for accessible tumors), somatostatin receptor ligands (e.g., octreotide, lanreotide), cabergoline, and pegvisomant (a GH receptor antagonist) [S4]. Surgery is effective in many but fails to achieve remission in patients with large or invasive macroadenomas [S4].

Diabetes mellitus is a common complication due to GH-induced insulin resistance, and patients have increased cardiovascular risk [S3].

Exam traps

- Confusing acromegaly with gigantism: gigantism occurs in children before epiphyseal closure, while acromegaly develops in adults after growth plates fuse [S1].
- Assuming all patients with acromegaly have diabetes: while ~1/3 develop diabetes, it is not universal [S3].
- Believing acromegaly is rare: it is rare but incidence may be rising in the elderly due to increased life expectancy [S5].
- Overestimating cancer risk: while long-term GH exposure may promote tumor development, epidemiological evidence remains inconclusive, and the association is debated [S6].

Common confusions

- Acromegaly is not caused by excess GH alone; it is defined by hypersecretion of GH and elevated IGF-1 [S1].
- GH-induced insulin resistance occurs independently of obesity and is paradoxically linked to low body fat [S3].

- The disease is not limited to young adults: elderly patients are increasingly diagnosed, and comorbidities are more prevalent in this group [S5].
- Treatment success is not guaranteed: even after surgery, many patients with macroadenomas remain non-remitted [S4].

10 retrieval questions

1. What is the primary cause of acromegaly? [S1][S4]
2. Which metabolic complication is most commonly associated with acromegaly and occurs in approximately one-third of patients? [S3]
3. How does growth hormone impair insulin signaling in acromegaly? [S3]
4. What imaging modality is recommended for identifying the underlying pituitary adenoma in acromegaly? [S4]
5. What is the first-line treatment for acromegaly in patients with accessible pituitary adenomas? [S4]
6. Why is early diagnosis crucial in acromegaly? [S1][S4]
7. What is the typical age range for diagnosis of acromegaly in younger patients? [S5]
8. What are the independent predictors of mortality in acromegaly according to current evidence? [S6]
9. How does the risk of diabetes mellitus in acromegaly differ from that in typical type 2 diabetes? [S3]
10. What is the current status of evidence regarding the link between acromegaly and cancer development? [S6]

Five-minute recap

Acromegaly results from GH hypersecretion due to pituitary adenoma, leading to elevated IGF-1 and systemic complications [S1][S4]. GH directly causes insulin resistance and metabolic dysfunction, even in lean patients [S3]. Diagnosis relies on biochemical testing and MRI [S4]. Treatment begins with surgery, followed by medical therapy if needed [S4]. Diabetes is common and carries cardiovascular risk [S3]. Mortality is linked to age, GH levels, and IGF-1 at diagnosis [S6]. The disease is increasingly seen in the elderly, and delayed diagnosis remains a key challenge [S5].

Educational study aid — verify important claims against the cited sources.

References

- [S1] PubMed PMID 11571931 — Acromegaly. <https://pubmed.ncbi.nlm.nih.gov/11571931/>
- [S2] PubMed PMID 9429856 — Acromegaly. <https://pubmed.ncbi.nlm.nih.gov/9429856/>
- [S3] PubMed PMID 38844688 — Diabetes mellitus in patients with acromegaly: pathophysiology, clinical challenges and management. <https://pubmed.ncbi.nlm.nih.gov/38844688/>
- [S4] PubMed PMID 35120696 — Diagnosis and Treatment of Acromegaly: An Update. <https://pubmed.ncbi.nlm.nih.gov/35120696/>
- [S5] PubMed PMID 31939489 — Acromegaly in the elderly patient. <https://pubmed.ncbi.nlm.nih.gov/31939489/>
- [S6] PubMed PMID 38007382 — Acromegaly and Cancer: An Update. <https://pubmed.ncbi.nlm.nih.gov/38007382/>